# Day 2 Recap

# Questions

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# Recap questions

1) Write the R code to assign the value 100 to an object called “hundred”
2) Write the R code to assign the values Monday, Tuesday, and Wednesday to an object called “worst_weekdays”
3) Write the R code to calculate the mean of the numbers 1–10

# Answers

1) Write the R code to assign the value 100 to an object called “hundred”

```
hundred <- 100
```

# Answers

2) Write the R code to assign the values Monday, Tuesday, and Wednesday to an object called “worst_weekdays”

**worst_weekdays <- c(“Monday”, “Tuesday”, “Wednesday”)**

# Answers

3) Write the R code to calculate the mean of the numbers 1–10

```
mean(c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
```

```
numbers <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
mean(numbers)
```

# Questions?

# Data Types and Structures

# Session Objectives

**By the end of this session, you'll be able to:**

# Data Types and Structures

- This session will cover how data is organized in R.

- Knowing how your data is organized and being able to adjust that organization to match your research is an important part of coding in R and data management.

# Data Types and Structures

For example:

- You might need your data in a certain structure to run statistical analyses or create visualizations.
- You may need to transform variables into types that are more easily understood by R.

# Data Types and Structures

There are two main ways R does this:

- **Data types:**
  - The kind of value that is in a cell/group of cells
- **Data structures:**
  - How values are organizes

# Data Types

# Data Types

- Values can come in many different formats and/or units.
    - Numbers with decimals: 13.094
    - TRUE/FALSE
    - Categorical names

- Knowing what type of data you have, and making sure that R knows what kind of data you have, is a foundational aspect of coding.

# Main data types in R

- **Numeric:**
  - Data values that consist of numbers. These numbers can be integers (whole numbers), or have decimal values.
- **Character:**
  - Data values that consist of letters or words. We played with some of these in Block 4, and we had to wrap these values in quotation marks to tell R that these are values and not functions.
- **Logical:**
  - Data values that are TRUE or FALSE. We can also ask R to provide TRUE or FALSE responses if a certain condition is met.

# Main data types in R

- **Factor:**
  - Data values that represent categories, and can store either numeric or character data. “Education level” would be a factor variable consisting of character data, with values like “High school diploma”, “Undergraduate degree”, “Masters degree”, etc. Likert scales are very common examples where numeric data has categorical representation, such as the number 1 meaning “Strongly disagree” and the number 5 meaning “Strongly agree”.

# Main data types in R

- **Dates and Times:**
  - Data values stored in date/time format. This can be tricky depending on how your raw/original data looks before it is brought in to R, but the format for dates can be something like “2026-05-06”, and time can look like “12:45:54”.

# Let’s jump into R!

# Data Structures

# Data structures

- Now that we've seen how R handles types of data values, we can now start looking at how those values collectively form larger **data structures.**

# Main data structures in R

- **Vectors:**
  - Vectors are the simplest data structure. **A vector is a list of elements of the same data type.**
  - For example, you could have a numeric vector containing the elements (10, 15, 20, 25) or a character vector containing the elements (“blue”, “red”, “green”).
  - A single variable in a dataset (e.g., a list of ages of all the participants) can be considered a vector.

# Main data structures in R

- **Data frames:**
  - Data frames are the most common data structure used in R. Data frames are what you see as spreadsheets.
  - Unlike vectors, which have one dimension (a single column of the same data type), data frames are two-dimensional, and have columns and rows that can store different data types in the same data frame (e.g., numeric and character).
  - **Data frames are made up of vectors of the same length.**

# Main data structures in R

- **Data frames do have some constraints:**
  - The columns and rows must be named
  - The lists of vectors in the data frame must have equal lengths (i.e., each column needs an identical number of elements/rows).
  - All elements in a single column must be of the same data type (e.g., you can’t switch between numbers and names in a single column).

# Main data structures in R

In addition to these two main data structures, there are also other data structures in R, which include lists and matrices. We will not focus on these structures in this course, but you can read more about data structures in the link on the webpage.

# Let’s jump back into R!